Static SR validation vs FreeSASA
0
20000
40000
60000
80000
100000
120000
FreeSASA SASA
zsasa f64
64 points
n=4,370
R²=1.000000
mean err=2.079e-05%
max err=0.0002209%
zsasa f64 SASA
zsasa f32
64 points
n=4,370
R²=1.000000
mean err=0.0001077%
max err=0.02799%
zsasa f32 SASA
zsasa bitmask f64
64 points
n=4,370
R²=0.999755
mean err=0.7575%
max err=2.413%
zsasa bitmask f64 SASA
zsasa bitmask f32
64 points
n=4,370
R²=0.999755
mean err=0.7575%
max err=2.413%
zsasa bitmask f32 SASA
zsasa bitmask corrected f64
64 points
n=4,370
R²=0.999748
mean err=0.7771%
max err=2.42%
zsasa bitmask corrected f64 SASA
zsasa bitmask corrected f32
64 points
n=4,370
R²=0.999747
mean err=0.7772%
max err=2.449%
zsasa bitmask corrected f32 SASA
RustSASA
64 points
n=4,370
R²=0.999935
mean err=0.4226%
max err=2.789%
RustSASA SASA
Lahuta
64 points
n=4,370
R²=0.999935
mean err=0.4227%
max err=2.789%
Lahuta SASA
Lahuta bitmask
64 points
n=4,370
R²=0.999701
mean err=0.841%
max err=3.04%
Lahuta bitmask SASA
zsasa f64
128 points
n=4,370
R²=1.000000
mean err=2.098e-05%
max err=0.0002267%
zsasa f32
128 points
n=4,370
R²=1.000000
mean err=0.0001012%
max err=0.01535%
zsasa bitmask f64
128 points
n=4,370
R²=0.999811
mean err=0.6618%
max err=2.024%
zsasa bitmask f32
128 points
n=4,370
R²=0.999811
mean err=0.6617%
max err=2.024%
zsasa bitmask corrected f64
128 points
n=4,370
R²=0.999716
mean err=0.8578%
max err=2.231%
zsasa bitmask corrected f32
128 points
n=4,370
R²=0.999716
mean err=0.8579%
max err=2.248%
RustSASA
128 points
n=4,370
R²=0.999973
mean err=0.2752%
max err=2.149%
Lahuta
128 points
n=4,370
R²=0.999973
mean err=0.2751%
max err=2.149%
Lahuta bitmask
128 points
n=4,370
R²=0.999768
mean err=0.7316%
max err=2.471%
zsasa f64
256 points
n=4,370
R²=1.000000
mean err=2.078e-05%
max err=0.0002568%
zsasa f32
256 points
n=4,370
R²=1.000000
mean err=8.653e-05%
max err=0.007903%
zsasa bitmask f64
256 points
n=4,370
R²=0.999789
mean err=0.7187%
max err=1.662%
zsasa bitmask f32
256 points
n=4,370
R²=0.999789
mean err=0.7186%
max err=1.662%
zsasa bitmask corrected f64
256 points
n=4,370
R²=0.999763
mean err=0.7939%
max err=1.634%
zsasa bitmask corrected f32
256 points
n=4,370
R²=0.999763
mean err=0.7941%
max err=1.634%
RustSASA
256 points
n=4,370
R²=0.999992
mean err=0.154%
max err=1.026%
Lahuta
256 points
n=4,370
R²=0.999992
mean err=0.154%
max err=1.026%
Lahuta bitmask
256 points
n=4,370
R²=0.999777
mean err=0.7367%
max err=1.771%
zsasa f64
512 points
n=4,370
R²=1.000000
mean err=2.085e-05%
max err=0.0002447%
zsasa f32
512 points
n=4,370
R²=1.000000
mean err=9.606e-05%
max err=0.005176%
zsasa bitmask f64
512 points
n=4,370
R²=0.999784
mean err=0.7404%
max err=1.313%
zsasa bitmask f32
512 points
n=4,370
R²=0.999784
mean err=0.7402%
max err=1.311%
zsasa bitmask corrected f64
512 points
n=4,370
R²=0.999777
mean err=0.7716%
max err=1.491%
zsasa bitmask corrected f32
512 points
n=4,370
R²=0.999777
mean err=0.7718%
max err=1.491%
RustSASA
512 points
n=4,370
R²=0.999997
mean err=0.0896%
max err=0.6818%
Lahuta
512 points
n=4,370
R²=0.999997
mean err=0.08973%
max err=0.6844%
Lahuta bitmask
512 points
n=4,370
R²=0.999997
mean err=0.08973%
max err=0.6844%
zsasa f64
1024 points
n=4,370
R²=1.000000
mean err=2.07e-05%
max err=0.0002054%
zsasa f32
1024 points
n=4,370
R²=1.000000
mean err=9.642e-05%
max err=0.002874%
zsasa bitmask f64
1024 points
n=4,370
R²=0.999788
mean err=0.7321%
max err=1.083%
zsasa bitmask f32
1024 points
n=4,370
R²=0.999788
mean err=0.7319%
max err=1.083%
zsasa bitmask corrected f64
1024 points
n=4,370
R²=0.999776
mean err=0.78%
max err=1.177%
zsasa bitmask corrected f32
1024 points
n=4,370
R²=0.999776
mean err=0.7802%
max err=1.177%
RustSASA
1024 points
n=4,370
R²=0.999999
mean err=0.05486%
max err=0.3612%
Lahuta
1024 points
n=4,370
R²=0.999999
mean err=0.0549%
max err=0.36%
Lahuta bitmask
1024 points
n=4,370
R²=0.999999
mean err=0.0549%
max err=0.36%